CUSTOMER SEGMENTATION AND CLUSTER ANALYSIS ON RETAIL CUSTOMER BEHAVIOR

Author: Aaheri Gangopadhyay

Repository: github.com/aaherigangopadhyaystats-code

ABSTRACT

Understanding customer diversity is key to targeted retail marketing. In this project, I analysed 200 mall shopper profiles (n = 200) using unsupervised learning techniques to group customers by demographic traits, income levels, and spending patterns. By evaluating univariate distributions, correlation structures, and multi-variable feature scaling, I built and compared multiple clustering approaches—including K-Means, Agglomerative Hierarchical Clustering, and density-based methods. Using the Elbow Method across inertia metrics, I identified an optimal 5-cluster solution in two-dimensional income-spending space, neatly separating shoppers into distinct behavioral segments like high-earning low-spenders and high-earning high-spenders. Scaling features with Z-score standardization further allowed 4-dimensional clustering across Age, Income, Spending Score, and Gender. These results offer clear, actionable profiles for tailored marketing strategies.

1. PROJECT OVERVIEW AND RATIONALE

Retail businesses often struggle to engage customers effectively when treating their entire base as a single homogenous group. By grouping shoppers who share similar income levels and spending habits, businesses can design tailored marketing campaigns that directly match customer priorities.

I designed this project to answer three practical questions:

- Do income and spending habits naturally group customers into clear, identifiable segments?
- What is the optimal number of clusters (K) when balancing mathematical compactness against business interpretability?
- How do feature scaling and non-numerical attributes (like Gender) change cluster boundaries when moving from 2D space to multi-attribute clustering?

2. EXPLORING THE DATASET

I worked with the Mall Customers dataset containing 200 customer profiles across 5 primary variables:

Customer ID: Unique identification string (dropped during analysis to prevent index bias).

Gender: Categorical feature (56% Female, 44% Male).

Age: Continuous variable ranging from 18 to 70 years (Mean = 38.85, Standard Deviation = 13.97).

Annual Income (k$): Continuous income ranging from $15k to $137k (Mean = $60.56k).

Spending Score (1-100): Synthetic metric assigned by the mall based on customer behavior and purchase patterns (Mean = 50.20).

Dataset Breakdown:

Observations: 200 rows, 0 missing values

Gender Ratio: 112 Females (56.0%), 88 Males (44.0%)

Peak Visitor Age Group: 32 years old (highest footfall frequency)

Key preliminary insights from exploratory data analysis include:

- Gender Averages: Female shoppers averaged a slightly higher spending score (51.53) than male shoppers (48.51), despite males having a slightly higher average income ($62.23k vs $59.25k).
- Feature Correlations: Age showed a moderate negative correlation with Spending Score (r = -0.327), indicating younger customers tend to spend more relative to older demographics. Annual Income and Spending Score showed almost zero direct linear correlation (r = 0.0099), confirming that spending habits cannot be predicted by income alone without clustering.

3. PREPROCESSING AND FEATURE PIPELINE

To prepare the data for distance-based clustering algorithms:

- Feature Cleanup: Dropped Customer ID as it carries no statistical signal.
- One-Hot Encoding: Converted the categorical Gender column into numerical binary format (Gender-Male).
- Feature Standardization (Z-Score Scaling): Distance metrics like Euclidean distance are heavily biased by feature magnitude (e.g., income values up to 137 vs. gender flags between 0 and 1). For multi-attribute clustering, I applied StandardScaler to ensure all features contributed equally:
- X_scaled = (X - Mean) / Standard_Deviation

4. CLUSTERING EXPERIMENTS AND FINDINGS

Experiment 1: 1D Income Clustering

I started by testing univariate K-Means on Annual Income alone with K = 3.

- Low Income Cluster: Mean income $33.49k (n = 74).
- Moderate Income Cluster: Mean income $69.75k (n = 104).
- High Income Cluster: Mean income $108.18k (n = 22).
- Inertia: 23,589.31

My Observation: While 1D income clustering separates salary tiers, it completely misses how customers actually choose to spend their money.

Experiment 2: 2D Bivariate Clustering (Income vs. Spending Score)

To capture real spending behavior, I ran K-Means across Annual Income and Spending Score. Using the Elbow Method across K = 1 to 10, I evaluated within-cluster sum-of-squares (inertia).

Elbow Method Inertia Trajectory:

K = 1 : 269,981.3

K = 2 : 181,363.5

K = 3 : 106,147.1

K = 4 : 73,679.7

K = 5 : 44,448.5 (Optimal Elbow Point)

K = 6 : 37,233.7

The 5 Customer Segments Identified (K = 5):

- Target / High Spenders (Cluster 1): High Income (Mean = $86.54k), High Spending Score (Mean = 82.13). Young average age (32.7 years).
- Careful / Conservative (Cluster 2): High Income (Mean = $88.20k), Low Spending Score (Mean = 17.11). Average age 41.1 years.
- Sensible / Moderate (Cluster 0): Moderate Income (Mean = $55.30k), Moderate Spending Score (Mean = 49.52). Average age 42.7 years.
- Careless / Spendthrift (Cluster 4): Low Income (Mean = $25.73k), High Spending Score (Mean = 79.36). Youngest average age (25.3 years).
- Low-Budget / Cautious (Cluster 3): Low Income (Mean = $26.30k), Low Spending Score (Mean = 20.91). Oldest average age (45.2 years).

Experiment 3: Multivariate Clustering and Alternative Algorithms

I expanded the input space to 4 dimensions (Age, Annual Income, Spending Score, Gender-Male) on scaled data and compared K-Means against Agglomerative Hierarchical Clustering.

- Multivariate K-Means: An elbow test on 4D standardized data showed a clear bend around K = 5, demonstrating that 5 primary groups remain stable even when age and gender are factored in.
- Agglomerative Hierarchical Clustering: Using Ward's linkage with K = 5, the hierarchical tree generated cluster boundaries closely mirroring the 2D bivariate segments, confirming the natural structure of the data.

5. MY TAKEAWAYS AND NEXT STEPS

- Distinct Spending Profiles: Income alone does not predict spending. The 2D K-Means model successfully separated high earners into two distinct groups—those who spend freely and those who hold back.
- Clear Target Audience: Cluster 1 (High Income, High Spending) represents the prime audience for luxury campaigns, while Cluster 4 (Low Income, High Spending) responds strongly to trend-focused promotional offers.
- Future Extension: Next, I plan to explore DBSCAN to test if density-based techniques can better handle border points, and calculate Silhouette Scores to measure cluster separation mathematically.